Abstract

This project describes the design, construction and implementation of an interactive hardware-generated laser shooting system which combines traditional analog timing circuits with IoT features. The system consists of two main components – an analog laser transmitter and a digital sensor-target receiver. The transmitter is actually a 555 timer integrated circuit in monostable firing mode, that generates ballistic "bullet" pulses.

The target module uses the ESP32 microcontroller with a high sensitivity Light Dependent Resistor (LDR) and a voltage divider circuit to sense laser attacks. If the hit is successful, the system gives immediate haptic and audio feedback by an integrated buzzer. The device uses the dual-core processor and Wi-Fi stack of the ESP32 to act as a local web server which pushes scoring information and timer to a web front-end application. It is a seamless integration of the old electronic parts and the new web-based telemetry which provides a scalable system for low cost, low latency recreational gaming and competitive target practice systems.

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1 Introduction

The use of interactive shooting and target games has grown in popularity both in the entertainment and educational sectors. The purpose of this project is to create a low cost and efficient laser shooting system using the mixture of simple analog electronic with the modern IoT technology.

The system to be implemented is comprised of two sections: laser transmitter and target detection unit. This transmitter is built with a 555 - timer integrated circuit, in monostable mode, with the laser ON for short pulses, rather than continuously. This provides a more realistic shooting experience to that of a single bullet firing mechanism.

On the receiving end, the signal is received by an ESP32 microcontroller and a Light Dependent Resistor (LDR) detects the incoming laser light. If the target is struck successfully, the buzzer sounds and the score is automatically displayed. The score and system data can also be viewed in real time via a local web interface as the ESP32 features Wi-Fi built-in.

A primary objective of this project is to show some of the possibilities of traditional electronic components like the 555 - timer combined with the new technology of embedded systems and wireless communication. The project offers a platform for playing, learning embedded systems and experimenting with interactive devices using IoT, which is simple, scalable and affordable.

In general, this system shows the practical application of analog circuits, sensor - based detection as well as wireless communications in a fun and engaging application.

1.1 Motivation

This project is motivated by the increasing research interest in interactive electronic game system and smart embedded technologies. The commercial laser shooting and target practice systems are also costly and they often utilize advanced hardware and software which are not easy to learn or build for students and hobbyists. It was developed to provide a simple, inexpensive and educational solution with readily available electronic components.

A second reason was to gain an understanding of the important role that traditional analog electronics can continue to serve in contemporary systems. The use of a 555 timer in monostable mode is an example of what can be achieved by using a simple timing circuit to give a realistic pulse based firing action, without the need for complete reliance on complex programming. Concurrently, the ESP32 microcontroller brings the latest in IoT capabilities, including real-time score monitoring and wireless communication.

The project also seeks to deliver a hands-on learning experience in fields like circuit design, sensor integration, embedded systems, and web-based monitoring. The system utilizes these technologies in a single interactive platform, making it a fun and educational learning tool for students fascinated with electronics and IoT development.

The overall motivation of this project is the design of an affordable, engaging and technically meaningful system which showcases the successful integration of analog electronics and modern smart electronics technologies.

1.2 Microcontroller

- ESP-32 Dev Kit

1.2.1 Sensor

- Light Dependent Resistor (LDR)

1.2.2 Figures of Microcontroller and Sensor

Figure 1.1 ESP-32 Dev Kit [1]

Figure 1.2 LDR [2]

1.2.3 References

The ESP-32 Dev Kit is used as the main controller unit in the system [1], while the Light Dependent Resistor (LDR) is utilized for laser detection and sensing purposes [2].

1.3 Chapter Summary

This chapter introduced the overall concept and background of the proposed laser shooting system. It discussed the motivation behind developing a low-cost interactive laser target platform by integrating analog electronics with IoT technology. The chapter also briefly introduced the main hardware components used in the project, including the ESP32 microcontroller and the Light Dependent Resistor (LDR) sensor. Overall, this chapter provides the foundational understanding and objectives necessary for the following design and implementation chapters.

2 Literature Review

Laser shooting and laser tag systems have become popular in recreational gaming and interactive training applications because they provide safe and engaging gameplay experiences. Modern laser tag systems commonly use infrared or laser-based transmitters together with sensor-based target detection mechanisms for hit registration [3]. Recent developments have also introduced wireless communication and real-time score monitoring features using embedded systems and web-based interfaces [4]. Furthermore, microcontrollers such as the ESP32 are widely used in interactive gaming systems because of their built-in Wi-Fi capability and real-time processing performance [5]. Inspired by these existing technologies, the proposed project integrates a 555 timer-based pulse firing mechanism with an ESP32 and LDR-based target detection system to create a low-cost interactive laser shooting platform.

2.1 Background Study

Interactive laser shooting systems are widely used in gaming, simulation, and recreational training applications. These systems typically consist of a laser transmitter, a target detection unit, and a feedback mechanism for score indication or hit confirmation. Traditional laser tag systems mainly rely on infrared or laser-based communication combined with embedded controllers for processing user interaction.

The development of low-cost microcontrollers and IoT technologies has significantly improved the design of modern interactive systems. Devices such as the ESP32 provide wireless communication, real-time processing, and web-based monitoring features, making them suitable for smart gaming applications. Similarly, analog electronic components such as the 555 timer IC continue to be widely used for pulse generation and timing applications because of their simplicity and reliability.

In this project, a 555 - timer configured in monostable mode is used to create single-shot laser pulses, while an ESP32 and LDR-based sensor module are used for target detection and score processing. The combination of these technologies creates an affordable and interactive laser shooting platform that integrates both analog and digital electronic systems.

2.1.1 An IoT Based Laser Shooting Game using ESP-32 Dev Kit and LDR Sensors

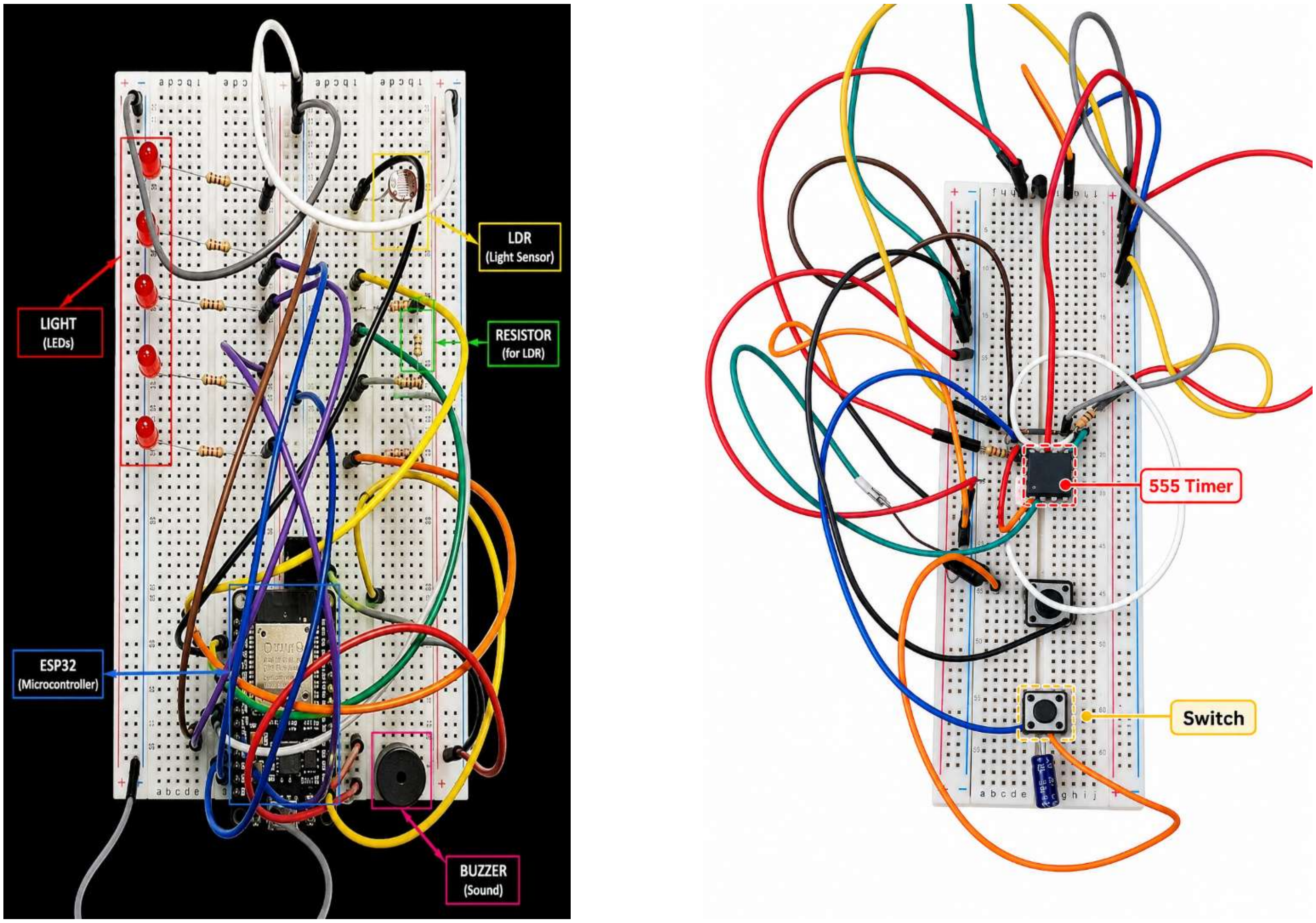

Figure 2.1: An IoT Based Laser Shooting Game using ESP-32 Dev Kit and LDR Sensors and Monostable gun switch

2.2 Chapter Summary

This chapter presented the background study and literature review related to interactive laser shooting and embedded IoT-based gaming systems. Existing technologies and concepts relevant to laser-based target detection, ESP32 microcontrollers, and sensor integration were briefly discussed

3 System Design

The proposed laser shooting system is designed by integrating an analog laser transmitter with a digital IoT-based target detection unit. The complete system consists of two main sections: the laser gun (transmitter) and the target module (receiver). The transmitter section uses a 555 timer IC configured in monostable mode to generate single-shot laser pulses when the trigger button is pressed. This creates a controlled pulse-based firing mechanism instead of continuous laser emission.

The receiver section is developed using an ESP32 microcontroller and an LDR sensor for laser detection. When the laser beam strikes the LDR sensor, the ESP32 processes the sensor data and activates feedback components such as LEDs and a buzzer. The ESP32 also utilizes its built-in Wi-Fi capability to transmit real-time score and hit information to a web-based monitoring interface.

The system is powered through a USB supply and designed using low-cost and easily available electronic components. The overall architecture ensures simple implementation, real-time responsiveness, and easy scalability for future improvements such as multiplayer functionality, wireless synchronization, and advanced scoring mechanisms.

3.1 Pin Defination

In the Figure 3.1, the pin definition of ESP-32 Dev Kit [1] is shown in the Table 3.1 a detailed pin description is given.

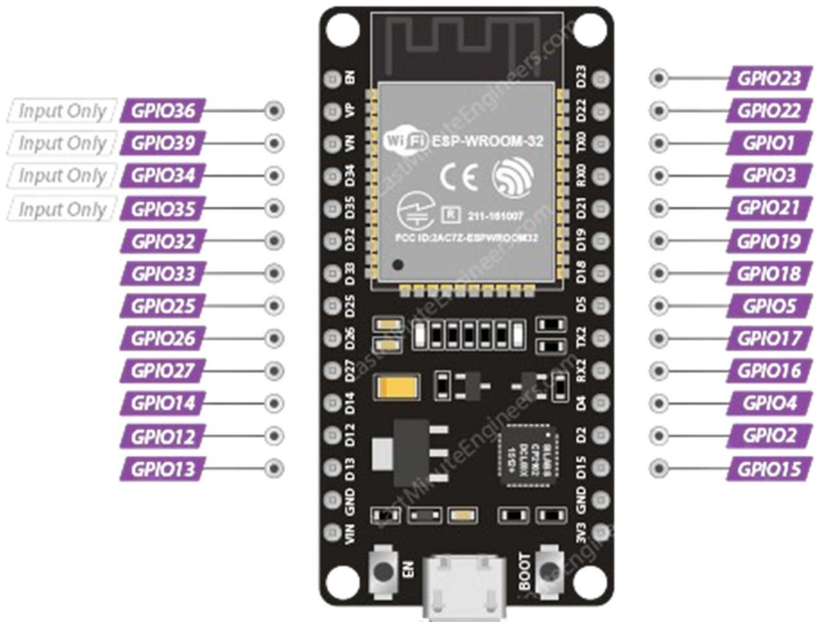

Figure 3.1: Pin Definition of ESP-32 Dev Kit

In the Figure 3.2, the pin definition of 555 timer [3] is shown in the Table 3.3 Pin description is also given.

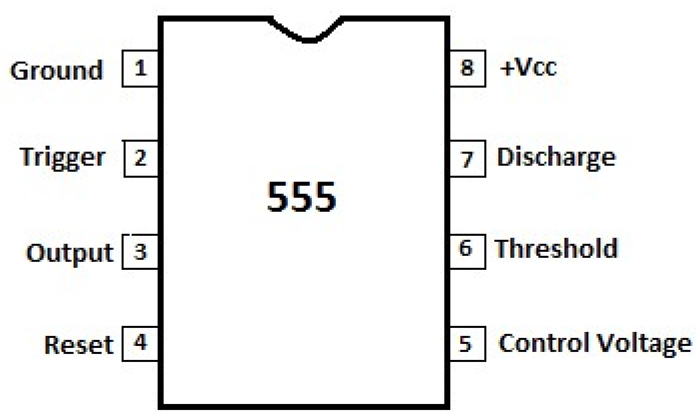

Figure 3.2: Pin Definition of 555 Timer

Pin Name	Type	Function
VIN	Power	External power input for ESP32 board (5V supply).
3V3	Power	3.3V regulated output pin.
GND	Power	Ground connection pin.
EN	Control	Enable/Reset pin for ESP32.
GPIO36 (VP)	Input	Analog input pin (ADC1), input only.
GPIO39 (VN)	Input	Analog input pin (ADC1), input only.
GPIO34	Input	GPIO pin with analog input capability, input only.
GPIO35	Input	GPIO pin with analog input capability, input only.
GPIO32	I/O	General-purpose input/output with ADC and PWM support.
GPIO33	I/O	General-purpose input/output with ADC and PWM support.
GPIO25	I/O	GPIO with DAC, ADC, and PWM functionality.
GPIO26	I/O	GPIO with DAC, ADC, and PWM functionality.
GPIO27	I/O	General-purpose input/output pin with ADC support.
GPIO14	I/O	Digital I/O pin supporting SPI communication.
GPIO12	I/O	Digital I/O pin with SPI and ADC capability.
GPIO13	I/O	Digital I/O pin supporting SPI and PWM.
GPIO23	I/O	SPI MOSI pin and general-purpose I/O.
GPIO22	I/O	Default I2C SCL pin and digital I/O.
GPIO1 (TX0)	Output	UART transmit pin for serial communication.
GPIO3 (RX0)	Input	UART receive pin for serial communication.
GPIO21	I/O	Default I2C SDA pin and digital I/O.
GPIO19	I/O	SPI MISO pin and general-purpose I/O.
GPIO18	I/O	SPI clock pin and digital I/O.
GPIO5	I/O	SPI chip select pin and digital I/O.
GPIO17	I/O	UART2 transmit pin and digital I/O.
GPIO16	I/O	UART2 receive pin and digital I/O.
GPIO4	I/O	Digital I/O pin with ADC and PWM support.
GPIO2	I/O	General-purpose digital I/O pin.
GPIO15	I/O	SPI and general-purpose digital I/O pin.
D0	Power	Connected internally for flash memory operation.
D1	Power	Connected internally for flash memory operation.
D2	Power	Connected internally for flash memory operation.
D3	Power	Connected internally for flash memory operation.
CMD	Power	Flash communication control pin.

Pin Name	Type	Function
CLK	Power	Flash clock signal pin.

Table 3.1: Pin Description of ESP-32 Dev Kit

Parameter	Description
Microcontroller	ESP32-WROOM-32
Operating Voltage	3.3V
Input Voltage	5V (USB/VIN)
Clock Frequency	Up to 240 MHz
Wi-Fi	802.11 b/g/n
Bluetooth	Bluetooth v4.2 and BLE
Flash Memory	4 MB
SRAM	520 KB
GPIO Pins	30+ GPIO Pins
ADC Channels	18 Channels
DAC Channels	2 Channels
PWM Support	Available
Communication Protocols	UART, SPI, I2C
USB Interface	Micro USB
Operating Temperature	−40°C to 85°C
Power Consumption	Low Power Mode Supported
Processor Type	Dual-Core Xtensa LX6
USB Programming	Supported
Wireless Range	Up to 100 m (Approx.)

Table 3.2: Parameters of ESP-32 Dev Kit

Pin No.	Pin Name	Description
1	GND (Ground)	Connected to the negative terminal or ground of the power supply.
2	TRIG (Trigger)	Starts the timing cycle when the voltage drops below 1/3 of the supply voltage.
3	OUT (Output)	Gives the output signal (HIGH or LOW) to drive LEDs, buzzers, relays, etc.

Pin No.	Pin Name	Description
4	RESET	Resets the timer when connected to LOW. Usually connected to VCC if not used.
5	CTRL (Control Voltage)	Used to control the timing internally. Normally connected to ground through a 0.01 µF capacitor to reduce noise.
6	THR (Threshold)	Stops the timing cycle when voltage rises above 2/3 of the supply voltage.
7	DISCH (Discharge)	Discharges the external timing capacitor during operation.
8	VCC	Connected to the positive power supply voltage.

Table 3.3: Pin Description of 555 Timer

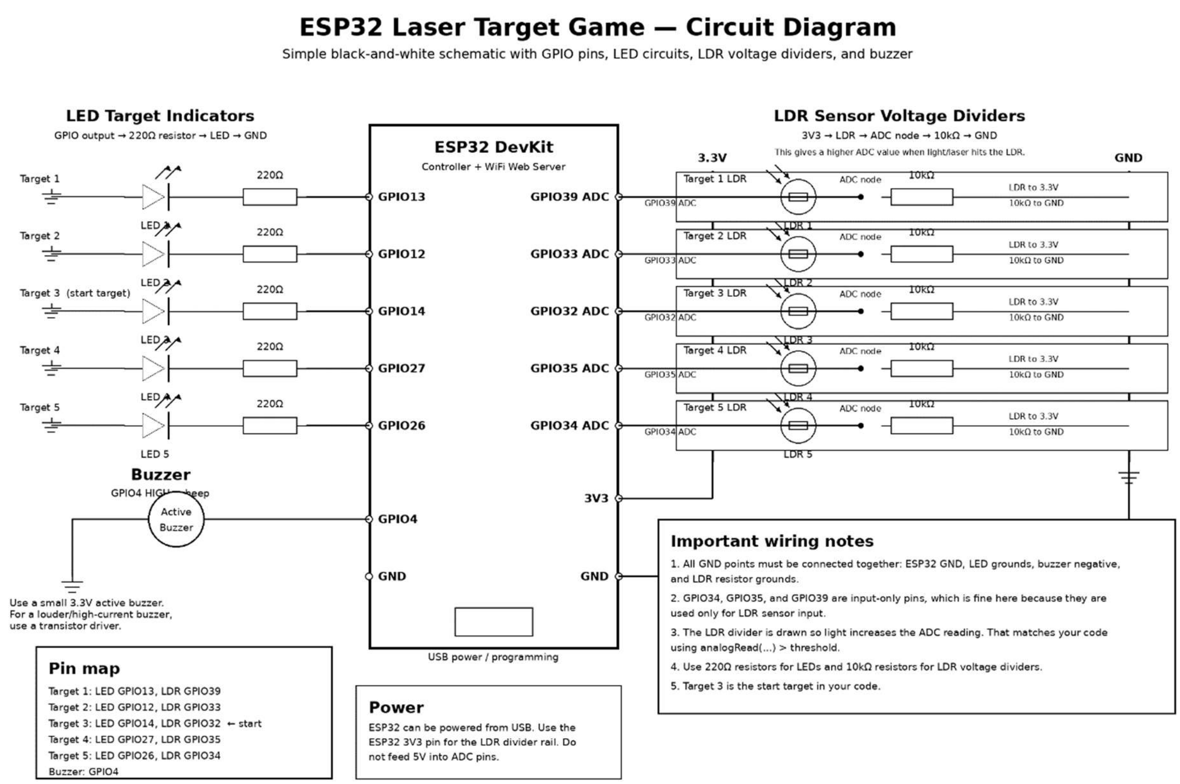

Figure 3.3: Circuit Diagram of LED Shooting Board

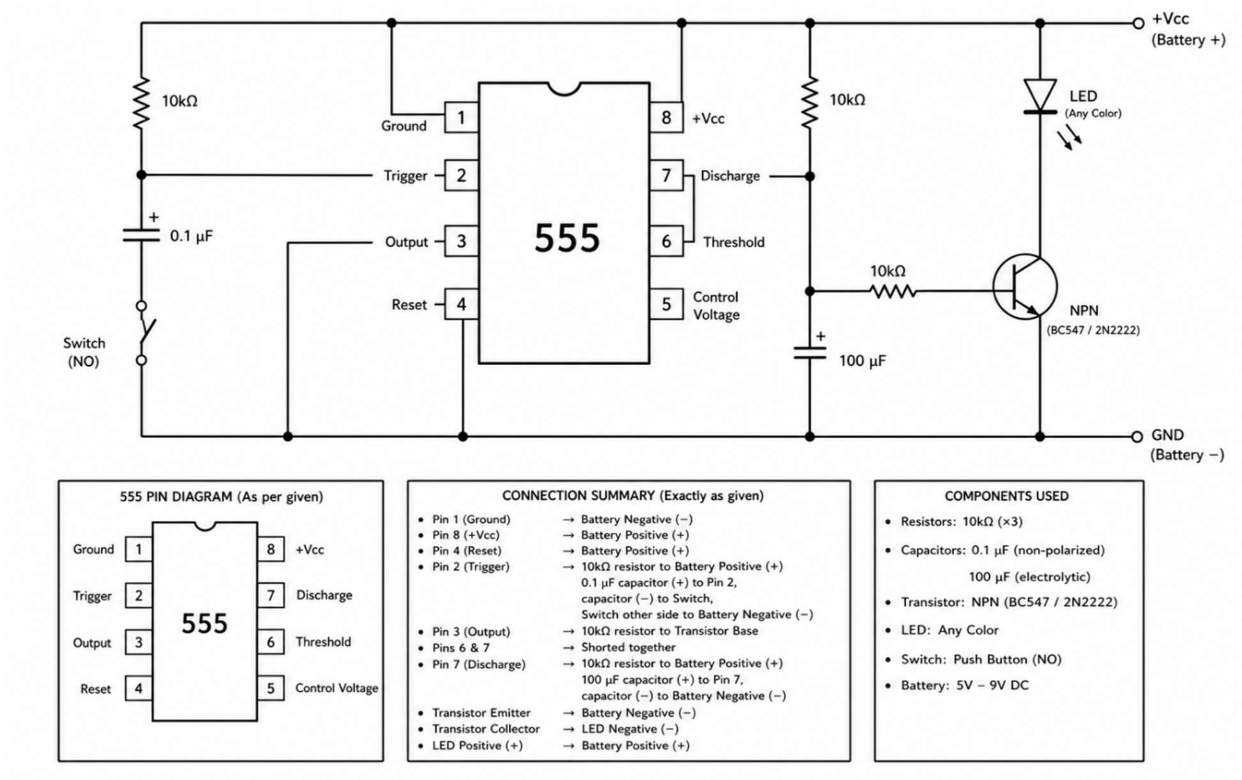

Figure 3.4: Circuit Diagram of Laser Shooter

3.2 Chapter Summary

This chapter presented the system design and hardware configuration of the proposed laser shooting system. The architecture and operational workflow of the system were discussed along with the major hardware components used in the implementation. In addition, the ESP32 microcontroller pin description and parameter specifications were provided to explain the functionality and technical characteristics of the controller unit used in the project. Overall, this chapter establishes the hardware foundation and design structure of the proposed system.

4 Implementation

The implementation phase of the proposed laser shooting system focuses on integrating both the transmitter and receiver modules into a functional interactive platform. The hardware components were assembled using a breadboard-based prototype design for easy testing and modification.

The laser transmitter section was implemented using a 555 timer IC configured in monostable mode. A push button was used as the trigger mechanism to generate short laser pulses through the laser module. This configuration allowed the system to simulate single-shot firing behavior instead of continuous laser output.

The receiver section was implemented using the ESP32 microcontroller along with an LDR sensor for laser detection. The sensor readings were continuously monitored by the ESP32 through its analog input pins. When the laser beam struck the LDR sensor, the ESP32 processed the signal and activated the buzzer and LED indicators to provide immediate feedback.

The ESP32 was programmed using the Arduino IDE environment and powered through a USB connection. Additionally, the built-in Wi-Fi capability of the ESP32 was utilized to support future web-based score monitoring and IoT integration features.

The complete system was tested under different lighting and shooting conditions to ensure stable operation and reliable target detection performance.

4.1 Implementation

- LDR connected with D32-D35.
- Buzzer connected with D4.
- Light connected with D12-D14, D27.
- 555 Timer connected with switch.

4.1.1 Configuration Code

Configuration code is given below.

```
#include <WiFi.h>

const char* ssid = "wifi_name";
const char* password = "wifi_pass";
WiFiServer server(80);
```

```
const int ledPins[5] = {13, 12, 14, 27, 26};
const int ldrPins[5] = {39, 33, 32, 35, 34};
const int buzzerPin = 4;

int threshold[5] = {2600, 2000, 2000, 2000, 2000};
unsigned long gameDuration = 120000;   // 2 minutes
unsigned long targetMaxTime = 3000;    // 3 seconds per target

void setup() {
  Serial.begin(115200);

  for (int i = 0; i < 5; i++) {
    pinMode(ledPins[i], OUTPUT);
    digitalWrite(ledPins[i], LOW);
  }

  for (int i = 0; i < 5; i++) {
    pinMode(ldrPins[i], INPUT);
  }
  pinMode(buzzerPin, OUTPUT);
  digitalWrite(buzzerPin, LOW);
  WiFi.begin(ssid, password);

  Serial.print("Connecting to Wi-Fi");
  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
  }

  Serial.println();
  Serial.println("Wi-Fi Connected");
  Serial.print("ESP32 IP Address: ");
  Serial.println(WiFi.localIP());
  server.begin();
  Serial.println("System Configuration Completed");
}
```

4.2 Chapter Summary

This chapter discussed the implementation and software configuration of the proposed laser shooting system. The hardware integration process, including the transmitter and receiver modules, was explained along with the functionality of the ESP32 microcontroller, LDR sensors, LEDs, and buzzer. The chapter also presented the ESP32 configuration and programming logic used for target detection, score processing, timing control, and Wi-Fi communication.

5 Conclusion

The proposed laser shooting system successfully demonstrates the integration of analog electronics with modern IoT-based embedded technology. The project combines a 555 timer-based monostable laser transmitter with an ESP32-powered target detection and scoring system to create an interactive and low-cost gaming platform.

The implementation verified that the LDR sensor and ESP32 microcontroller can effectively detect laser impacts and provide real-time feedback through LEDs, buzzer alerts, and score processing mechanisms. The use of Wi-Fi capability further enhances the system by enabling future web-based monitoring and multiplayer expansion possibilities.

One of the major achievements of this project is the successful combination of traditional electronic components with modern wireless embedded systems in a practical and engaging application. The system provides both educational and recreational value while maintaining simple hardware architecture and low implementation cost.

Overall, the project demonstrates the feasibility of developing an interactive smart laser shooting platform using affordable and easily available components, making it suitable for learning, experimentation, and future research in embedded and IoT-based gaming systems.

5.1 Limitations

- Sensor accuracy may be affected by ambient light.
- Limited number of targets supported.
- Breadboard setup is less stable.
- Manual sensor threshold adjustment is required.
- Wi-Fi communication is limited to local networks.
- No multiplayer support in current version.
- No cloud or mobile application integration.
- Prototype is not power optimized.

5.2 Future Work

- Develop a custom PCB for improved stability and compact design.
- Add multiplayer and wireless synchronization features.
- Integrate mobile application support for score monitoring.
- Introduce advanced game modes and scoring systems.

References

[1] Espressif Systems, “ESP32 DevKit,” Espressif, 2025. [Online]. Available: https://www.espressif.com/. [Accessed: 8-May-2026].

[2] Components101, “LDR Sensor,” Components101, 2025. [Online]. Available: https://components101.com/. [Accessed: 8-May-2026].

[3] H. Berlin, “The 555 Timer Applications Sourcebook,” Howard W. Sams & Co., 1984.

[4] Espressif Systems, “ESP32 Series Datasheet,” Espressif Systems, 2025. [Online]. Available: https://www.espressif.com/.

[5] Components101, “LDR Sensor Tutorial and Applications,” Components101, 2025. [Online]. Available: https://components101.com/.